

Boutiquaat Studio AI Hub

Complete Technical Architecture, Application & DevOps Documentation

Enterprise Production v1.0.0

LIVE DEPLOYMENT URL	GITHUB REPOSITORY	VERCEL PROJECT	TARGET BRANCH
boutiquaat-gen-app.vercel.app	bagzmax7/boutiquaat-gen-app	maxlumagas-projects	main

1. Executive Summary & Core Platform Overview

Boutiquaat Studio AI Hub (`boutiquaat-gen-app`) is an enterprise-grade AI visual synthesis and workflow orchestration application built specifically for e-commerce, digital brand marketing, and creative production pipelines. The platform unifies next-generation generative AI infrastructure—spanning AI image generation, text-to-video, image-to-video, batch background removal, automated social media canvas resizing, and specialized AI creative agents—into a unified Next.js dashboard backed by Supabase database state management and RunningHub OpenAPI v2 execution engines.

2. Technology Stack & Framework Architecture

Frontend Technologies

- **Framework & Architecture:** Next.js 14+ with App Router (React Server & Client Components).
- **Language & Type Safety:** TypeScript 5.0+ enforcing strict type interfaces across components and API responses.
- **UI Styling & Iconography:** Tailwind CSS with custom glassmorphism design system, Lucide React icons, and custom CSS animations.
- **Client State & Hooks:** Custom React Context providers and specialized hooks (`useTasks` for polling and database sync, `useAuth` for session governance).
- **UI Feedback & Media Handling:** React Hot Toast for instant feedback, React Dropzone for file uploads, client-side video/image previews.

Backend & Serverless Infrastructure

- **Serverless Handlers:** Node.js runtime running on Vercel Serverless Functions (`app/api/*`).
- **Database & Authentication:** Supabase PostgreSQL (managing tables: `tasks` , `users` , `workflows` , and `image_agent_sessions`) with custom middleware JWT protection (`middleware.ts`).
- **AI API Infrastructure (RunningHub OpenAPI v2):**
 - **AI App Execution:** `POST /openapi/v2/run/ai-app/{appId}`
 - **ComfyUI Workflow Engine:** `POST /openapi/v2/run/workflow/{workflowId}`
 - **Standard Image API:** Models under `rhart-image-*` namespace.

- **Standard Video API:** Multi-provider endpoints under `bytedance/*`, `rhart-video/*`, `kling-*`, and `google/veo3.1-*`.
- **Media CDN Binary Uploads:** POST `https://www.runninghub.cn/openapi/v2/media/upload/binary`
- **RunningHub LLM API:** OpenAI-compatible completions endpoint `https://llm.runninghub.ai/v1/chat/completions`.

3. Comprehensive Application Features & Modules

1. Boutiquaat Video AI Studio (`/video`)

A specialized video generation studio supporting top-tier model endpoints with intelligent parameter sanitization:

- **Supported Model Endpoints:**
 - `bytedance/seedance-2.0-global/image-to-video` (Seedance 2.0 Global)
 - `seedance-2.0-global-fast/image-to-video` (Seedance 2.0 Fast)
 - `rhart-video/sparkvideo-2.0/text-to-video` (SparkVideo 2.0 Text-to-Video)
 - `rhart-video/sparkvideo-2.0/image-to-video` (SparkVideo 2.0 Image-to-Video)
 - `rhart-video/sparkvideo-2.0/multimodal-video` (SparkVideo 2.0 Multimodal)
 - `kling-video-o1/image-to-video` & `kling-v3.0-std-image-to-video`
 - `google/veo3.1-pro/start-end-to-video-channel-low-price`
- **Engine Parameter Validation:**
 - **Resolution Mapping:** Maps `resolution` and `quality` (e.g. `720p`, `1080p`) simultaneously to satisfy API schemas.
 - **Duration Sanitization:** Automatically converts UI duration strings (e.g. `"10s"`) into integer values (`10`) within the valid API range (4–15 seconds).
 - **Payload Cleansing:** Filters out empty array fields (`imageUrls: []`) to prevent API schema validation errors.
 - **Reference Image Validation:** Enforces mandatory image upload for image-to-video models before sending requests.

2. AI Image Agent Hub (`/image-agent`)

A conversational AI assistant tailored for commercial creative generation:

- **Specialized Roles:** Concept Architect, Lifestyle Brand Manager, Ad Creative Specialist, Anime Illustrator.
- **Multi-Turn Vision & Chat:** Supports text prompting, reference image uploads, LLM-based prompt enhancement, and session persistence in Supabase.

3. Social Resize & PSD Layer Studio (`/studio/social-resize`)

Multi-platform canvas editor supporting automated asset adaptation for Instagram Story (9:16), Feed (1:1), Landscape (16:9), and Banner dimensions with cross-origin proxy image handling (`/api/proxy-image`).

4. Batch Background Removal Modules

- **Batch Image BG Removal:** Parallel ComfyUI background extraction for e-commerce catalog photos.
- **Batch Video BG Removal:** Video green-screen/background stripping yielding transparent or studio-lit MP4 output files.

5. ComfyUI Workflow Bundling & Auto-Analysis (`/bundling`)

Scans workflow JSON schemas to extract editable node fields, optimize API key channel selection, and execute batch tasks.

6. Real-Time Task Manager & Admin Operations (`/tasks` , `/history` , `/admin/tasks`)

- **Task Polling Hook (`useTasks`):** Real-time background polling for task statuses (`QUEUED` , `RUNNING` , `SUCCESS` , `FAILED`).
- **Analytics & Costs:** Displays execution time, coin consumption, and money spent per task.
- **Admin Control Dashboard:** Allows enterprise vs. consumer API key channel switching and task re-execution.

4. GitHub Source Control & Repository Workflow

The project codebase is hosted on GitHub under `https://github.com/bagzmax7/boutiqaat-gen-app`. Continuous integration and automated deployments are driven by Git branch pushes to `main`.

Git Configuration & Strict Author Enforcement

⚠️ CRITICAL: Vercel Author Email Requirement

The Git commit author email **MUST strictly match** `bagzmax21@gmail.com` (the owner email of the Vercel account). If a commit uses a different email address, Vercel will block the deployment with error `seatBlock: COMMIT_AUTHOR_REQUIRED`.

Mandatory Local Git Configuration Commands:

```
# Set correct Git author credentials globally or locally
git config user.email "bagzmax21@gmail.com"
git config user.name "bagzmax7"

# Verify configuration:
git config user.email # Output MUST be: bagzmax21@gmail.com
```

Fixing Author Email on Past Commits (if blocked):

```
git commit --amend --reset-author --no-edit
git push --force-with-lease origin main
```

Standard Development & Commit Pipeline

1. **Local Development Verification:** Verify the production build compiles cleanly before committing:

```
npm run build
```

2. **Staging Files:** Stage modifications (excluding secrets and generated artifacts specified in `.gitignore`):

```
git add .
```

3. **Committing Changes:** Follow standard conventional commit formatting:

```
git commit -m "feat: add sparkvideo 2.0 endpoints and duration validation"
```

4. **Pushing to Main:** Push to GitHub to automatically trigger the Vercel deployment build:

```
git push origin main
```

5. Vercel Deployment & Production Infrastructure

Production hosting is managed on Vercel via direct GitHub Git Integration.

- **Live Production URL:** `https://boutiqaat-gen-app.vercel.app`
- **Vercel Dashboard:** `https://vercel.com/maxlumagas-projects/boutiqaat-gen-app`

Automated Build Pipeline

When code is pushed to `main`, Vercel executes the following build sequence:

1. Clones the repository state from GitHub.
2. Installs dependencies via `npm install`.
3. Executes the build command: `next build`.
4. Generates Serverless Functions for all API route handlers under `/api/*` and static pages.
5. Deploys to the production edge network and updates the DNS record for `boutiqaat-gen-app.vercel.app`.

Environment Variable Configuration (Vercel Settings)

All secret keys and API credentials are configured in Vercel Settings → Environment Variables.

Secrets are intentionally excluded from Git repository tracking.

Environment Variable	Description	Target Environment
NEXT_PUBLIC_SUPABASE_URL	Supabase project URL	Production / Preview / Dev
NEXT_PUBLIC_SUPABASE_ANON_KEY	Supabase public client key	Production / Preview / Dev
SUPABASE_SERVICE_ROLE_KEY	Supabase service role secret key	Production / Serverless Only
RUNNINGHUB_API_KEY_ENTERPRISE	RunningHub Enterprise API Key	Production / Serverless Only
RUNNINGHUB_API_KEY_CONSUMER	RunningHub Consumer API Key	Production / Serverless Only
RUNNINGHUB_BASE_URL	https://www.runninghub.ai	Production / Serverless Only
RUNNINGHUB_UPLOAD_URL	https://www.runninghub.cn	Production / Serverless Only
AUTH_USERNAME	Admin authentication username	Production / Serverless Only
AUTH_PASSWORD	Admin authentication password	Production / Serverless Only
AUTH_SECRET	JWT token signature secret	Production / Serverless Only
GEMINI_API_KEY	Google Gemini Vision API key	Production / Serverless Only
HF_TOKEN	Hugging Face Vision token	Production / Serverless Only

Triggering Manual & Empty Redeploys

If environment variables are updated in Vercel or a build needs to be re-triggered without code changes:

```
# Push an empty commit to trigger a new Vercel build
git commit --allow-empty -m "chore: trigger vercel redeploy"
git push origin main
```

6. Supabase Database Administration & Schema Maintenance

- **Supabase Dashboard:** <https://supabase.com/dashboard/project/aszmkbwliopbybawqco>
- **SQL Editor:** <https://supabase.com/dashboard/project/aszmkbwliopbybawqco/sql/new>

Schema Synchronization Guidelines

1. When creating new database tables or columns, apply the DDL SQL statements in the Supabase SQL Editor.
2. Update the local tracking file `supabase/schema.sql` to reflect the latest database structure.
3. Commit and push `supabase/schema.sql` to GitHub to maintain schema version history.

7. Comprehensive API Endpoint Reference

API Endpoint Path	HTTP Method	Request Summary / Description
/api/runninghub/video-generate	POST	Validates and dispatches video generation requests to RunningHub OpenAPI v2.
/api/runninghub/image-generate	POST	Dispatches image generation requests (rhart models).
/api/runninghub/upload	POST	Uploads binary image/video references to RunningHub CDN.
/api/runninghub/query	POST	Queries task status, execution cost usage, and output URLs.
/api/image-agent/sessions	GET / POST	Retrieves and creates AI Image Agent conversation sessions.
/api/image-agent/enhance	POST	Refines user prompts using RunningHub LLM completions.
/api/tasks	GET / POST	Reads and updates task execution states in Supabase.
/api/proxy-image	GET	CORS proxy stream for downloading external media assets safely.